Data Science – Midterm Project Report Guidelines

This is midterm the project guidelines. Its main part is statistical inference: get inferences from your data using hypothesis testing. Know your tests and the reason for using them for your problem, and conduct a comprehensive comparison with other tests.

The report should contain the following:

1) Where you have collected the dataset, how many rows and columns there are, and whether #instances × #attributes is at least 3M after encoding (show the arithmetic). Name some features with their interpretation to the real world.
2) How did you prepare your data (encoding, outliers, missing values, duplicates), and why do you follow such steps?
3) What are the changes you got after data preparation? Give before-and-after observations.
4) What are your basic questions about your data, and how is your problem statement novel from them? Show a few EDA plots, each with a one-line takeaway.
5) Your statistical inferences: conduct 3–5 hypothesis tests on your problem statement(s). For each test:
    a) Before running the test, write the null and alternative hypotheses (H_0 carries the equality), the significance level, one- or two-sided, the test you will use, and the smallest difference that would matter in the real world.
    b) Which other tests did you consider, and why did you not use them?
    c) Report the test statistic, degrees of freedom, p-value and decision, and the estimate with a 95% confidence interval. Is the difference large enough to matter?
6) The report should contain link to codes written by you and a readme file.
7) Submit the report as a PDF of at most 8 pages by 27 October 2026, 11:59 pm, on the classroom. You have to use the LaTeX report template.
8) This will be followed by the midterm project presentation whose guidelines will be shared soon.

In general for this task, you will be graded based on how much you could dig into and infer from the data, not on whether a test rejects H_0.

All the best!